Bioavailability of herbs and spices in humans as determined by ex vivo inflammatory suppression and DNA strand breaks.

OBJECTIVE: The aim of this work was to determine the bioavailability of herbs and spices after human consumption by measuring the ability to protect lymphocytes from an oxidative injury and by examining the impact on inflammatory biomarkers in activated THP-1 cells. METHODS: Ten to 12 subjects in each of 13 groups consumed a defined amount of herb or spice for 7 days. Blood was drawn from subjects before consumption and 1 hour after taking the final herb or spice capsules. Subject serum and various extractions of the herbs and spices were analyzed for antioxidant capacity by oxygen radical absorbance capacity (ORAC) analysis or by 1,1-diphenyl-2-picrylhydrzyl (DPPH). Subject peripheral blood mononuclear cells (PBMCs) in medium with10% autologous serum were incubated with hydrogen peroxide to induce DNA strand breaks. Subject serum was also used to treat activated THP-1 cells to determine relative quantities of 3 inflammatory cytokine (tumor necrosis factor-α [TNF-α], interleukin-1α [IL-1α], and IL-6) mRNAs. RESULTS: Herbs and spices that protected PBMCs against DNA strand breaks were paprika, rosemary, ginger, heat-treated turmeric, sage, and cumin. Paprika also appeared to protect cells from normal apoptotic processes. Of the 3 cytokine mRNAs studied (TNF-α, IL-1α, and IL-6), TNF-α was the most sensitive responder to oxidized LDL-treated macrophages. Clove, ginger, rosemary, and turmeric were able to significantly reduce oxidized LDL-induced expression of TNF-α. Serum from those consuming ginger reduced all three inflammatory biomarkers. Ginger, rosemary, and turmeric showed protective capacity by both oxidative protection and inflammation measures. CONCLUSIONS: DNA strand breaks and inflammatory biomarkers are a good functional measure of a food's bioavailability.